Assignment 3 — Database Systems and Cybersecurity (Project Build, Part 1)

> **Notice.** Keep the same team and the same project as in Project Stage 1. Start in the first days of Week 6: installing a database tool, creating the schema and loading test data always takes longer than expected, and the security part needs a working database to protect. Agree inside the team who owns which table, which part of the prototype and which security measure — every member must be able to reproduce their own part live at the defense.

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1. Purpose

This assignment is the **first practical build stage of your final project**. In Project Stage 1 you designed the concept, the requirements and the architecture. Here you implement and protect the part of that architecture where the data lives: a working database for your project, and the security measures that keep its data safe.

The result is not a separate exercise — it becomes part of the system you demonstrate at **Project Stage 2 (End-term, Week 9)** and submit and defend as the **Final Project (Week 10)**. Assignment 4 continues the build with the smart-technology and data-analysis layer on top of the database you create now.

Required reading

- Lecture notes for Weeks 6–7 (LMS)
- Parsons, J. J. (2022). *Computer Concepts: Comprehensive*, 21st ed. — **Module 10: Databases** and **Module 7: Digital Security**
- Nurpeisova, T. B., & Kaidash, I. N. (2017). *Information and Communication Technologies*. Almaty: Bastau

> **Note.** Submission and defense take place in the **Week 6** practice session, as scheduled in the syllabus. The cybersecurity material of Part C is covered in the Week 7 lecture, so read Parsons Module 7 in advance; the lecture will deepen what you apply here.

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2. Tasks

Part A — Data model and working database (25 points)

A1. Entity-relationship diagram. Draw the ERD of your project data. Include at least **three entities** with their attributes, primary keys and foreign keys, and mark the cardinality of every relationship (one-to-one, one-to-many, many-to-many). Resolve any many-to-many relationship with a linking table. Tools: draw.io, dbdiagram.io, MySQL Workbench or paper photographed clearly.

A2. Normalisation. Show one table of your design **before and after normalisation** to third normal form, and explain in 80–120 words which anomaly each step removed (insertion, update or deletion anomaly).

A3. Implementation. Create the database in a real DBMS — SQLite (with DB Browser for SQLite), MySQL, PostgreSQL, or a cloud database such as Firebase Firestore or Supabase. Submit:

- the `CREATE TABLE` statements with data types and constraints (`PRIMARY KEY`, `FOREIGN KEY`, `NOT NULL`, `UNIQUE`, `CHECK`);
- a screenshot of the schema inside the tool;
- at least **100 rows of realistic test data** in the main table and at least **10–70 rows** in each other table.

Part B — DBMS choice and data management (15 points)

B1. Comparison. Compare **three database options** that could serve your project — for example SQLite, PostgreSQL and Firebase Firestore. Use criteria such as data model (relational or document), hosting (local or cloud), concurrent users supported, cost, ease of backup, and security features. Present it as a table.

B2. Justification. In 100–150 words, justify the option you chose against the requirements you wrote in Project Stage 1, Part B. Refer to at least two specific requirements.

B3. Backup and recovery. State where the data is physically stored, how often it will be backed up, where the backup is kept, and how you would restore the system after a failure. Attach evidence of **one backup you actually made** — an exported `.sql` dump, a copy of the database file, or an export from the cloud console.

Part C — Security of the project data (25 points)

C1. Threat model. Identify the assets your system holds (accounts, personal data, content, logs) and complete a table of at least **five realistic threats**:

#	Asset	Threat	Likelihood (low/medium/high)	Impact (low/medium/high)	Countermeasure
1	User accounts	Password reuse and credential stuffing	Medium	High	Password hashing, rate limiting, two-factor authentication
2	...	...	...	...	...

Cover at least three different categories, for example malware, phishing and social engineering, SQL injection, weak authentication, data interception, insider misuse, data loss and denial of service.

C2. Protection measures. Describe the measures your project applies, grouped as: authentication and authorisation (including how passwords are stored — never in plain text), input validation against SQL injection, protection of data in transit (HTTPS/TLS), access control roles, backups, and logging. Then **implement one measure and prove it**: a screenshot of hashed passwords in your table, a parameterised query in your code with a short explanation, a role-based permission setting in your cloud console, or an equivalent.

C3. Personal data and privacy. List every item of personal data your system collects and, for each, state why it is necessary, how long it is kept and who can access it. Remove at least one item you do not truly need and say why (data minimisation). Add 80–120 words on your responsibilities when handling other people's personal data, referring to the Kazakhstan law on personal data and its protection or an equivalent regulation.

Part D — Integration with the project (20 points)

D1. Working prototype. Show a **running prototype of your project with the database integrated** — not screenshots of the database alone. The prototype must connect to the database you built in Part A and perform at least **two operations through the interface**: one that reads data and displays it to the user, and one that writes data (creating, updating or deleting a record). Any technology is accepted: a web page, a mobile app, a desktop application, a command-line tool or a low-code platform. Submit the source code or the project link, a short note on how to run it, and screenshots of both operations working. The prototype will be extended in Assignment 4 and demonstrated again at Project Stage 2.

D2. Updated architecture. Redraw the architecture diagram from Project Stage 1 so that it now shows the database, where it is hosted, and the security boundary (what is encrypted, what is authenticated, what is public).

D3. Change log. A short table of what changed since Stage 1: requirements added, removed or reworded, design decisions revised, and the reason for each change. Three to six rows is enough.

D4. Contribution table. One row per team member stating which tables, prototype features and security measures that member personally produced.

Part E — Defense (15 points)

Present your work for **4–6 minutes** in the Week 6 practice session and answer **2–3 individual questions**. Every team member speaks about the part they built.

> **Note.** The instructor may ask you to **demonstrate the work live**: run the prototype and perform a read and a write in front of the instructor, open the database tool and show the schema or the hashed passwords, write a small query on request, or restore your backup. Bring a laptop with the database and the tools installed and working offline.

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3. Deliverables checklist

☐ Report with Parts A–D, exported as a single PDF named `Group_Team_ICT_Assignment3.pdf`

☐ Database file or export (`.sqlite`, `.sql` dump, or a link to the cloud project with access for the instructor)

☐ ERD and updated architecture diagram, readable at full page width

☐ Screenshots of the schema and the test data

☐ Prototype source code or project link, with instructions to run it, and screenshots of the read and write operations working

☐ Evidence of one implemented security measure and one backup

☐ Contribution table naming what each member built

☐ Reference list (any consistent style — APA, IEEE or the style used in your textbook)

☐ Uploaded to the LMS before the deadline; one submission per team, with all members named

☐ Every member ready to run their own part live

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4. Grading rubric (100 points)

Criterion	Excellent (90–100%)	Good (70–89%)	Satisfactory (50–69%)	Fail (0–49%)	Max
A. Data model and working database	Correct ERD with keys and cardinalities; normalisation explained accurately; database implemented with full constraints and the required volume of realistic test data	Model mostly correct with minor key or cardinality errors; database works; test data thinner than required	Simplistic model; few constraints; little or unrealistic test data	No working database; ERD missing or wrong	25
B. DBMS choice and data management	Thorough comparison; choice justified against specific Stage 1 requirements; credible backup plan with real evidence	Comparison present; justification partly generic; backup described with weak evidence	Superficial comparison; choice asserted; no real backup	No comparison or no justification	15
C. Security of the project data	Realistic threat model with sound ratings and matching countermeasures; measures described precisely; one measure genuinely implemented and proven; privacy analysis concrete and minimised	Threats mostly realistic; measures described but partly generic; implementation evidence thin	Few generic threats; countermeasures listed without connection to the system; no evidence	No threat analysis; passwords or data stored unsafely with no awareness of it	25
D. Integration with the project	Prototype runs and performs both a read and a write against the project database; architecture updated correctly; change log explains decisions; contributions clear and balanced	Prototype runs with one operation working or minor errors; updates present with small inconsistencies	Prototype incomplete or not connected to the database; diagram unchanged; vague contributions	No prototype; no integration with the project	20
E. Defense	Confident live demonstration; answers accurately, including a new question asked on the spot	Demonstrates the work; minor hesitation on details	Explains only part of the work; cannot modify anything live	Cannot run or explain own submitted work	15
				Total	**100**

Formatting, citation quality and readability are assessed inside criteria A–D. Work that cannot be defended orally is graded as the defense score alone, regardless of the written quality.

In team projects the score is awarded individually where contributions differ: a member who cannot explain or reproduce the part attributed to them in the contribution table receives a reduced mark for the corresponding criteria.

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5. Sample defense questions

1. Why is this attribute a foreign key, and what happens to the related rows if I delete this record?
2. Show me a table that is not in third normal form. What anomaly would it cause?
3. Write a query, right now, that counts the records of this table grouped by that column.
4. Why did you choose this DBMS instead of the other two you compared?
5. Your laptop is stolen tonight. What is lost, and how long would recovery take?
6. How are passwords stored in your database? Show me. Why is that safer than storing the text?
7. What exactly is SQL injection, and which line of your code prevents it?
8. Which personal data item did you decide not to collect, and why?